2021-22学年 春季学期 概率论与数理统计期中(回忆版)

Part One - single Choice

1. Assume A, B, C are three events, then $\bar{A}\bar{B}\bar{C}$ means that ( ).

(A) one of the events A, B, C happens;

(B) two of the events A, B, C happen;

(C) no more than one of the events A, B, C happen;

(D) none of the events A, B, C happens.

2. There are three people who are independently guessing a password. The probability of individually getting the password is 1/5, 1/3, 1/4 respectively. The probability of getting the password is ( ).

(A) 1/5 (B) 2/5 (C) 3/5 (D) 4/5

3. Assume random variables $X \sim N(0,4)$, $Y \sim N(1,4)$, and they are independent to each other. Then, $X-Y$ follows the distribution of ( ).

(A) $N(-1,0)$ (B) $N(-1,32)$ (C) $N(-1,8)$ (D) $N(1,8)$

4. Assume that random variables X and Y independent and identically distributed. The distribution of r.v. X is $F(x)$. The distribution function of $Z=\max(X,Y)$ is ( ).

(A) $F^2(z)$ (B) $1-F^2(z)$ (C) $(1-F(z))^2$ (D) $1-(1-F(z))^2$

5. Assume the distribution of r.v. X is $F(x)$, and $F(a)=0.5$. Which statement is true? ( )

(A) There is a point of $x_0<a$, which makes $F(x_0)>0.5$;

(B) There is a point x_0, which makes $F(x_0)>1$;

(C) Any point of $x>a$ makes $F(x)=1$;

(D) Any point of $x>a$ makes $F(x)\geq 0.5$.

Part Two - Blank Filling

1. Events A and B are independent to each other. Events C and A, Events C and B are disjoint events. Furthermore $P(A)=0.5$, $P(B)=0.4$, $P(C)=0.2$. Assume D is the event that at least one of the three Events A, B, C happens, thus $P(D)=$_____.

2. There are two events A, B, and they have $P(AB)=P(\bar{A}\bar{B})$. If $P(A)=p$, then $P(B)=$_____.

3. Assume there are two independent events A and B. The probability that both of them don't happen is 1/9. The probability of A happens and B doesn't is the same as the probability of B happens and A doesn't. Then $P(A)=$_____.

4. Assume that $F_1(x)$, $F_2(x)$ are one dimensional distribution with the constant $a,b>0$. If $a\cdot F_1(x)+b\cdot F_2(x)$ is a distribution, then the constants a, b satisfy the condition of _____.

5. Assume the random variable $X \sim N(-3, 25)$, and $Y = -2(X+3)$. Then, the distribution of r.v. Y is $Y \sim$ _____.

6. Assume the random variable Y follows exponential distribution with the parameter 1. If a is a constant being greater than 0, then $P\{Y \leq a+1 \mid Y > a\} =$ _____.

7. Assume there are two independent random variables X and Y, they follow normal distribution $N(0,1)$ and $N(1,1)$ respectively. Then $P\{X+Y \leq 1\} ==$ _____.

8. Assume that the joint density function of random variables (X, Y) is $f(x,y) = \begin{cases} \frac{1}{2}, & 0 \leq x \leq 1, 0 \leq y \leq 2 \\ 0, & \text{others} \end{cases}$ Then the probability that at least one of the events $\{X \leq 0.5\}$ and $\{Y \leq 0.5\}$ happens is _____.

9. Assume there are two independent random variables X and Y, they all follow uniform distribution $U[0,3]$. Then $P\{\max\{X, Y\} \leq 1\} =$ _____.

10. Assume that X and Y are two random variables, and $P\{X \geq 0, Y \geq 0\} = \frac{3}{7}$, $P\{X \geq 0\} = P\{Y \geq 0\} = \frac{4}{7}$. Then $P\{\max\{X, Y\} \geq 0\} =$ _____.

Part Three – Question Answering

1. A student needs to select 1 choice from 4 optional choices for answering a question. The student will randomly select a choice if having not got the answer, and of course will select it if having obtained the answer. Now the student's selected choice is the answer, what is the probability that the student has obtained the answer before selecting a choice based on the following se scenarios?

   a) The probability of having obtained the answer before selecting a choice is 0.5.

   b) The probability of having obtained the answer before selecting a choice is 0.2.

2. Assume the random variable X has $P\{X=1\} = P\{X=2\} = \frac{1}{2}$. Under the condition $X=i$, the random variable Y follows uniform distribution $U(0,i)$ $i=1,2$. What are the distribution function $F_Y(y)$ and the density function $f_Y(y)$?

3. If a hen lays eggs X which follows Poisson distribution with the parameter λ. The probability that each egg transforms into a chick is p. Determine the probability of each hen has k chicks $(Y=k)$ such that Y follows Poisson distribution with the parameter λp.

4. Assume $Y = X^2$, and the density function of r.v. X is $= f_X(x) = \begin{cases} cx, & 0 < x < 2 \\ 0, & \text{others.} \end{cases}$

   (a) What is the constant c?

   (b) What is the density function $f_Y(y)$?

5. Assume the distribution functions of r.v. X and r.v. Y are as follows.

$$F_X(x)=\begin{cases}0, & x<0,\\ \frac{1}{3}, & 0\le x<1,\\ 1, & x\ge 1,\end{cases} \qquad F_Y(y)=\begin{cases}0, & y<1,\\ \frac{1}{2}, & 1\le y<2,\\ 1, & y\ge 2.\end{cases}$$

Furthermore, $P(X=1,Y=1)=\frac{1}{3}$.

a) What is the joint frequency function of r.v. X and r.v. Y?

b) Are X and Y independent?

c) When $Y=1$, what is the conditional probability $P(X=k|Y=1)$?

6. Assume the joint density function of the two-dimensional random variable (X,Y).

$$f(x,y)=\begin{cases}ke^{-(x+y)}, & 0<x<1,\ 0<y<\infty,\\ 0, & \text{others}.\end{cases}$$

a) Compute the constant k;

b) Find the marginal density function of $f_X(x)$, $f_Y(y)$;

c) Find the distribution function of $Z=\max\{X,Y\}$.